

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IVth Semester of BSc Biotechnology/Biochemistry/Microbiology Examination- April- 2019

BS406: Genetic Engineering

Date: 27/4/2019 Day: Saturday Time: 1.30 p.m. To 2.00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose/write the correct answer for the following questions.

20

1. Find the correct arrangement in ascending order with respect to its foreign gene carrying capacity

- | | |
|---------------------------------|---------------------------------|
| (i) BAC-Plasmid-Cosmid-M13mp2 | (ii) M13mp2-cosmid-phagemid-BAC |
| (iii) BAC-Cosmid-Plasmid-M13Mp2 | (iv) Cosmid-BAC-Plasmid-M13mp2 |

2. Plasmid vectors used in cloning often contain a gene for the *N*-terminal 146 amino acids of the enzyme β -galactosidase. What is the purpose of including this gene in the vector

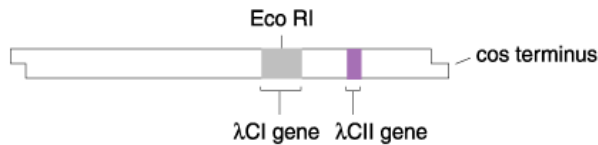
- | | |
|--|--|
| (i) It allows selection of <i>E. coli</i> host cells that contain the plasmid. | (ii) It allows selection of <i>E. coli</i> host cells that contain plasmid in which the insert has been ligated. |
| (iii) It cleaves the insert to allow it to be ligated into the vector | (iv) It enables the plasmid vector to replicate in <i>E. coli</i> host cells |

3. What is the natural function of restriction enzymes?

- | | |
|---|---|
| (i) Protecting bacteria by cleaving the DNA of infecting viruses. | (ii) Protecting bacteria by cleaving their own DNA. |
| (iii) Protecting bacteria by methylating their own DNA. | (iv) Protecting bacteria by methylating the DNA of infecting viruses. |

4. β - galactosidase is synthesized from _____ gene.

5.



What type of cloning vector does this diagram represent?

- (i) Insertion vectors (ii) Replacement vectors (iii) Cosmid (iv) All of above
6. Below are steps performed to make *E. coli* bacteria that can make human insulin protein for the health care market. Arrange them in proper order
- Use heat shock or electroporation to get the recombinant human insulin gene to go into a living *E. coli* cell.
 - Grow the *E. coli* in fermentation culture, extract and sell Humulin.
 - Modify the human gene and give it a promoter that will be turned on in *E. coli* cells.
 - Extract the DNA from a human who does not have diabetes Isolate and clone (make copies of) the gene encoding the insulin protein
- (i) a,b,c,d & e (ii) d,e,c,a&b (iii) b,c,d,a&e (iv) e,b,c,d & a
7. A student noted the following points regarding *Agrobacterium tumefaciens*:
- A. tumefaciens* is a gram-negative soil bacterium.
 - Opine catabolism genes are present in T- DNA region of Ti-plasmid.
 - Opines are synthesized by condensation of amino acids and α-ketoacids or amino acids and sugars.
 - A callus culture of crown gall tissue caused by *A. tumefaciens* in plants can be multiplied without adding phytohormones.
- Which one of the combinations of above statements is correct?
- (i) i, ii and iii (ii) i, ii and iv (iii) ii, iii and iv (iv) i, iii and iv
8. Ri plasmids are originally obtained from
- (i) *E. coli* (ii) *A. rhizogenes* (iii) *A. tumefaciens* (iv) *S. aureus*
9. If PCR is used to introduce random mutations rather than specific mutations, it is called as:
- (i) mutagenic PCR (ii) random PCR (iii) error prone PCR (iv) cloning PCR
10. A labelled DNA or RNA segment, which is used to find a specific sequence of nucleotides in a DNA molecule by hybridization is named as-
- (i) Vector (ii) Probe (iii) Primer (iv) Sense strand
11. Which of the following is considered as a reporter protein?
- (i) DNA ligase (ii) Xgal (iii) ampicillin (iv) GFP
12. Give the name for any one phagemid_____
13. The given DNA sequence (5' ATGC 3') is incubated with enzyme Terminal deoxynucleotidyl transferase and Adenine nucleotides, on which particular end the enzyme will add the adenine nucleotides?
- (i) Both 3' and 5' (ii) 3' (iii) 5' (iv) None
14. *E. coli* lysogenized with phage p22 is grown on nutrient agar medium. The culture plate was layered with top agar inoculated with λ phage. How does this superinfection affect the growth of *E. coli*, phage p22 or phage λ?
- (i) Due to *E. coli* cell lysis clear plaque (ii) Phage p22 does not allow

- will appeared on plate
- (iii) Phage λ inhibits the growth of phage p22
- (iv) superinfection by phage λ
Phage λ cannot grow on p22 lysogenized strain of *E. coli*
15. **Lambda phage vectors carry a copy of the lacZ' gene. Insertion of new DNA into this gene inactivates β -galactosidase synthesis. Recombinant plaques are distinguished by plating cells onto X-gal agar: plaques comprising non-recombinant phages are appeared as _____**
- (i) Clear (ii) Turbid (iii) Blue (iv) None
16. **From the following which is the major limitation of Agrobacterium derived vectors?**
- (i) *A. tumefaciens* and *A. rhizogenes* infect only dicotyledonous plants
(ii) *A. tumefaciens* and *A. rhizogenes* infect only monocot plants
(iii) There are very less restriction sites on natural Ti-Plasmid
(iv) None of above
17. If you want to clone a gene so that you may deduce the amino acid sequence of the protein from the DNA,library would be most useful to start from.
18. **In site-directed mutagenesis**
- (i) The primer contains the desired mutation
(ii) The template used for extension contain desired mutations
(iii) Mutated nucleotides are added in to reaction
(iv) All of above
19. **An undergraduate student experimenting on ligation of EcoRI fragments of λ gt10 and amylase gene, he had attempted all the preceding steps successfully. Wishing to select recombinants, he infected *E. coli* cells with possibly chimeric λ phage. Unfortunately, no clear or turbid plaque was observed. From the following which observation better explains the above-mentioned condition?**
- (i) λ gt10 does not carry any gene for amylase
(ii) No insertional inactivation of marker gene of λ gt10
(iii) *E. coli* is not an ideal host for λ gt10
(iv) The wild-type *E. coli* cell, in its native form does not support insertion of foreign DNA.
20. **A group of researchers involve in generating gene library for newly isolated bacteria. The organism's DNA contains numerous palindromic sequences, subsequently digested to approximately 20 KB sized fragments. Suggest the most appropriate vector system for cloning of these fragments in prokaryotic host.**
- (i) pBR322 (ii) pUC19 (iii) λ ZapII (iv) YIP

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
IVth Semester of BSc Biotechnology/Biochemistry/ Microbiology Examination- April- 2019

BS406: Genetic Engineering

Date: 27/4/2019 Day: Saturday Time: 2.00 p.m. To 4.30 p.m. Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q-IIA Answer the following questions as directed [Any Two] 10

- 1 Explain Ti plasmid and its various components?
- 2 Explain strategies for engineering
1)Co-integrate vector 2) Binary Vector
- 3 Which library will be used to study promoter and enhancer region of a gene?
Justify your answer.

Q-IIB Answer the following [Any one] 6

- 1 Explain following methods for gene transfer
[a] Gene gun [b] Electroporation
- 2 What is genomic DNA library? Explain two methods for screening of desired clone from a library

Q-IIC Write the answer for the following through pictorial diagram [Any One] 4

- 1 What is site-directed mutagenesis? Explain primer extension method for generating site-directed mutagenesis.
- 2 What is reverse transcription? Describe synthesis of cDNA.

SECTION - II

Q-IIIA Answer any Four of the following 20

- 1 On your trip to the Amazon you were introduced to a new plant that the indigenous people use as an anti-viral treatment. You took samples home to your lab and you found that this plant makes a protein (the PV protein) that prevents viral replication. Excited by the possible anti-AIDS applications, you construct a genomic DNA library from this plant in the hope of cloning the PV gene. a) What is a genomic DNA library? b) You get a DNA sample from cells, digest it with a restriction enzyme, and clone it into a vector. List 3 features of the vector that is absolutely required for your library construction.
- 2 Explain the functions of adaptor and linkers in molecular cloning.
- 3 What is restriction endonuclease (RE)? How does it differ from exonucleases. Discuss various types of Restriction endonuclease. Give the name of any two RE with acronym
- 4 What are the requirements of vector and insert with respect to ligation reaction? How enzyme alkaline phosphatase does useful in ligation of blunt end molecules?
- 5 Give a name for lysogenic bacteriophage? Discuss the plaque morphology for lysogenic? Discuss lambda replacement and insertion vectors in detail.

Q-IIIB Explain the selection strategies employed through following markers with name of representative vectors [Any Two] 10

- 1 red, gam
- 2 lacZ and lacZ'
- 3 CI, CII